

Enhanced Mobile App Security for Healthcare Applications

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Abstract—As the number of mobile device usage continues to rise, the security of mobile apps is becoming an increasingly pressing concern as they have become an integral part of our daily lives. Mobile application, particularly those handling sensitive data like healthcare information, face increasing threats from malicious actors seeking unauthorized access. This paper explores the integration of Advanced Encryption Standard (AES) for data encryption and Elliptic Curve Cryptography (ECC) for key encryption in a healthcare application. The paper demonstrates how AES-ECC encryption can be integrated into the core of the application, creating a fortress of confidentiality and integrity. AES encryption safeguards healthcare data from unauthorized access, ensuring its confidentiality and compliance with regulations like HIPAA while ECC's efficient key management allows for shorter encryption keys, enhancing app performance in mobile app environments with limited storage and computational power. The performance of the AES-ECC setup is also tested, comparing power consumption and CPU time against scenarios where only AES encryption is used. The findings contribute to the advancement of mobile app security practices and offer insights and guidelines for developers to safeguard sensitive data against evolving cyber threats.

Keywords—Security, AES, ECC, mobile apps, healthcare

I. INTRODUCTION

Security for mobile applications refers to the protection of data as it is transmitted across the networks via software run on mobile platforms like BlackBerry, iOS, Android, and Windows Mobile. Ensuring the security of mobile applications is essential in the modern linked world where mobile devices are our constant companions. The importance of safeguarding the apps they run, has increased due to the rapid spread of these devices and an increasing number of cyberattacks. It is important for users to view the security of mobile applications, which allow them to guarantee that the apps they download are secure, as well as raise market awareness of the presence of hackers and viruses used to steal the privacy and personal details of smartphone users [1]. This paper aims to identify the limitations of current security measure uses to secure mobile apps, to propose an algorithm to enhance security of mobile apps in healthcare and to evaluate the performance of the proposed techniques, the encryption algorithm.

The healthcare industry is the biggest and most rapidly expanding sector globally [2]. Medical information

like health data, health history, and medication can easily be accessible by patient through healthcare applications at any time and from any location. This can improve the quality of life for patients, increase cooperation, improve patient outcomes, reduce costs, and enhance overall efficiency of e-healthcare services. However, due to the fact that it is connected to an internal network that is linked to the Internet, it is vulnerable to various types of malicious attacks and viruses from devices.

Since Roman era, cryptography which is the art of secret writing is already used to hide information and making sure the message secure. A common technique for information security under cryptography is data encryption and decryption. Data encryption is the process of translating the data from plaintext (unencrypted) to ciphertext (encrypted) while data decryption is vice versa. Encryption is widely used to ensure data safety during transmission, especially for sensitive or confidential information. Encryption can also prevent data tampering, eavesdropping, and unauthorized access by malicious actors. Healthcare app often deal with personal or sensitive data that needs to be protected from unauthorized access or modification.

II. CURRENT SYSTEM

A. Cryptography

Since Roman era, cryptography which is the art of secret writing is already used to hide information and making sure the message secure. A common technique for information security under cryptography is data encryption and decryption. Data encryption is the process of translating the data from plaintext (unencrypted) to ciphertext (encrypted) while data decryption is vice versa. In order for the user to encrypt and decrypt the data, they need the encryption key and decryption key. The workings of the encryption and decryption is shown in Figure 2.1 below.

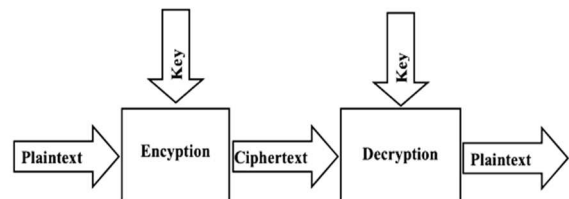


Figure 1: Encryption and decryption process [3]

The types of cryptography are symmetric and asymmetric algorithms. Symmetric algorithm provides secure communication between sender and receiver by using the same key for encryption and decryption meanwhile, asymmetric algorithm uses different pair of keys which are private and public key to encrypt and decrypt data. Key size which is the most important parameter for both symmetric and asymmetric is to be said larger in asymmetric. In other words, asymmetric provides more secure for more sensitive data [4]. Asymmetric is newer technology compared to symmetric.

1) Advanced Encryption Standard (AES)

Advanced Encryption Standard (AES) is one of the examples of symmetric algorithm. It is a secret-key algorithms or in other words, it uses the same key for encryption and decryption to secure communication [5]. Figure 2.2 below shows the architecture of the AES algorithm. AES is categorized as a block cipher and works with fixed-size data blocks. The AES technique encrypts for 128-bit keys, ten cycles are required for both encryption and decryption. To obtain the final encoded message, use 192-bit keys for 12 rounds and 256-bit keys for 14 rounds [6]. Features of AES include speed and coding density across a wide variety of formats, an easy-to-use interface, and protection against all known threats.

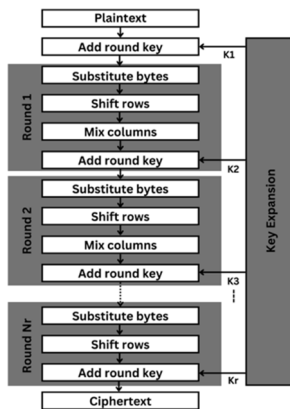


Figure 2: The architecture of AES algorithm [5]

2) Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography (ECC) is the example of asymmetric algorithm. It uses different keys for encoding and decoding. In 1985, V. Miller of IBM and N. Koblitz of the University of Washington created it. The algebraic structures of left-shaped carvings in certain domains served as the foundation for ECC. It works well enough to guarantee security when using a 164-bit key. For such system to be secure, a 1024-bit key is required. When bit sizes are the same, ECC offers the highest level of security. Due to its lower energy consumption, it is also beneficial for battery backup. The primary benefit of utilizing ECC is that, in comparison to other public key cryptosystems, it offers an equivalent level of security with a smaller key size, thereby minimizing processing cost and leads to rapid encoding also lower the energy consumption [6], [7]. It also provides solutions for non-repudiation, availability, authenticity, and secrecy of data [8].

B. Intrusion Detection System

Intrusion Detection System (IDS) is still essential for improving security in a variety of areas, including mobile apps. It remains as crucial part of cybersecurity tactics. The intrusion detection system (IDS) monitors on a host's or network's activities and notifies the system administrator of any intrusion it finds. Since Anderson (1980) published his seminal article about network security monitoring at the beginning of the 1980s, research concerning intrusion detection have been undertaken. As a result, the IDS has strengthened its position as a widely used defense tool for conventional IP networks, with a few available alternatives [9].

IDS systems fall into one of three categories which are behavior-based, signature-based, or both. Signature-based intrusion detection systems (IDS) compare incoming network traffic to a database of known attack patterns in order to identify potential threats. However, behavior-based intrusion detection systems monitor the typical behavior of people and systems and report any action that deviates from that pattern as potentially dangerous [8].

III. RELATED WORKS

To identify the limitation of current security measure uses to secure mobile app, a thorough assessment and analysis from various research methods is done to identify gaps in current knowledge and improve security. It compares their flaws and performances to develop new solution.

Montoya et al.'s 2013 study compared encryption algorithms for mobile applications, finding SERPENT offers highest security, TWOFISH for battery saving, and SERPENT is more demanding. However, SERPENT is heavier for smartphones and tablets due to its more complex structure and higher number of rounds compared to TWOFISH [3]. In 2016, Patel et al. proposed a hybrid method using ECC and Blowfish Algorithm for mobile cloud data security. This method reduced execution time but may impact resource-constrained devices [7]. Rajaprakash et al. introduced Rajaprakash Bagathbasha21 (RB21) in 2020, offering higher security but higher execution time due to its complexity[10].

Cinar et al. discuss malware detection techniques using signature-based and machine learning-based methods. Machine learning is more effective in detecting zero-day attacks, but has false positive and false negative risks, plus detection system cannot prevent the attack [11]. Hasan et al. propose cryptographic techniques for IoMT devices, but these may increase complexity and computational cost, impacting performance and efficiency[12]. Almalawi et al. propose the Lionized remora optimization-based serpent (LRO-S) encryption method for encrypting sensitive patient data before cloud storage. This hybrid metaheuristic optimization and improved security algorithm combines lion and remora fitness functions, enhancing medical professionals' access to cloud-based patient-sensitive data. The method minimizes execution time and improves privacy standards. However, since it is a new method, the security is less guaranteed [13]. Santis et al. discuss the use of ChaCha20 stream cipher and Poly1305 authenticator cryptographic algorithms for high-speed IoT applications

but it introduce a timing leak that could be exploited by an attacker [14].

Prabu et al. propose a modified ECC algorithm to avoid zero values and resist Side Channel Attacks, but it may introduce overheads [15]. Fayyaz et al. propose a secure IoMT system for long-term care, using the Advanced Encryption Standard (AES) algorithm for data collection, encryption, and transmission. However, AES is vulnerable to attacks like side-channel, differential, and key-sensitivity attacks, potentially compromising data security and privacy [16]. Kodzo et al. propose a two-level cryptographic technique using AES and ECC algorithms to improve data security in cloud computing, claiming it enhances confidentiality, integrity, and trust, but may not prevent brute-force attacks [17].

IV. PROPOSED WORK

Based on the review done on the previous article, the limitations and research gap are identified in safeguarding the data of mobile applications across the years using current security techniques which are cryptography and IDS. The proposed solution is to implement AES-ECC algorithm to secure a healthcare app. This is because AES algorithm is one of the most robust security protocols and approved by NIST. However, according to previous study, it is prone to side-channel attacks and we found out that ECC can fill in the limitation [18]. In 2020, Kodzo et al. has proposed the similar solution to use two-level cryptographic of AES and ECC algorithm. However, it poses a limitations like brute-force attack may not be prevented [17]. In this paper, we enhance out app by setting a limit to the number of failed login attempt. We set the limit to be not more than 3 times. This can strengthen the security and prevent from brute force attack. Figure 5 shows the application process flow.

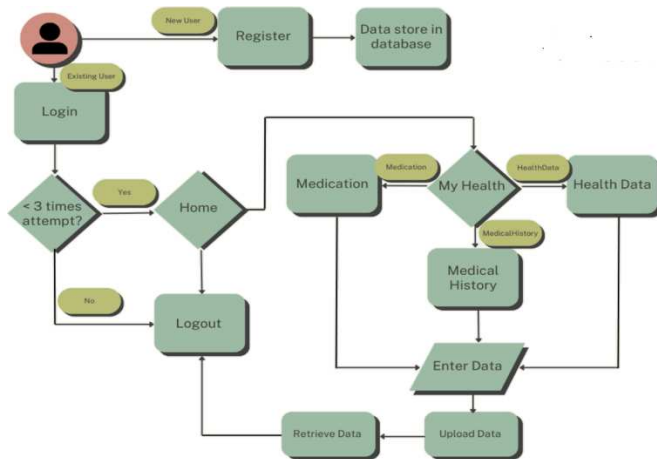


Figure 4: Application Process Flow

Another thing that distinguishes this work and the previous one is that it encrypts data using AES and it further encrypts the AES key using the ECC algorithm. Meanwhile the previous solution uses AES and ECC algorithm both to encrypt data. The advantage of this paper compared to the previous one is that it addresses the key management concerns more effectively. Figure 3.3 below

shows the implementation design of encryption and decryption.

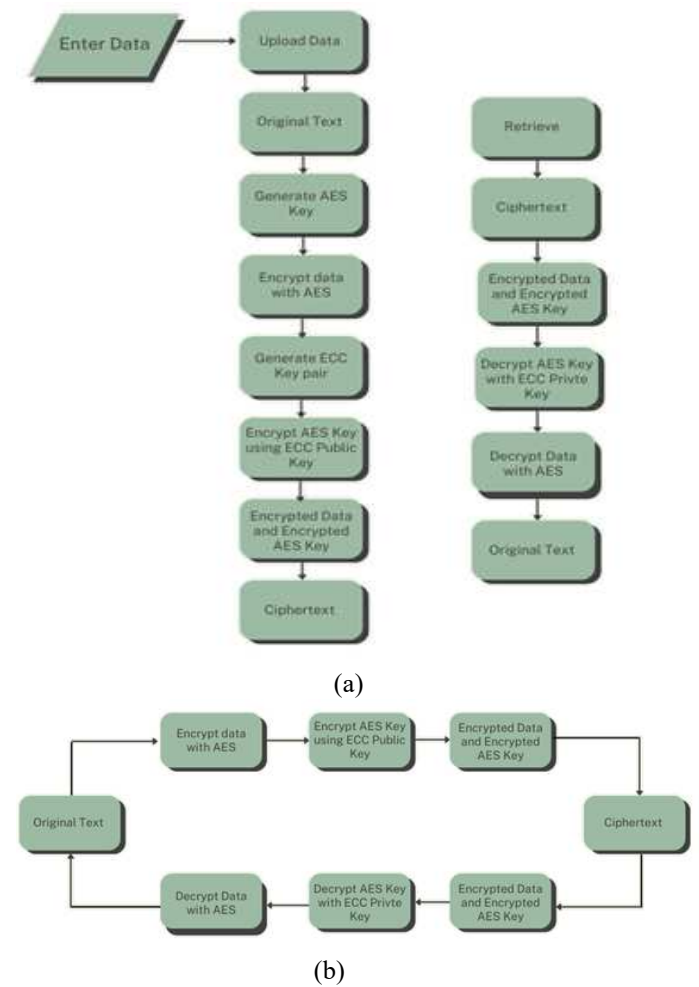


Figure 5: Encryption and Decryption Data

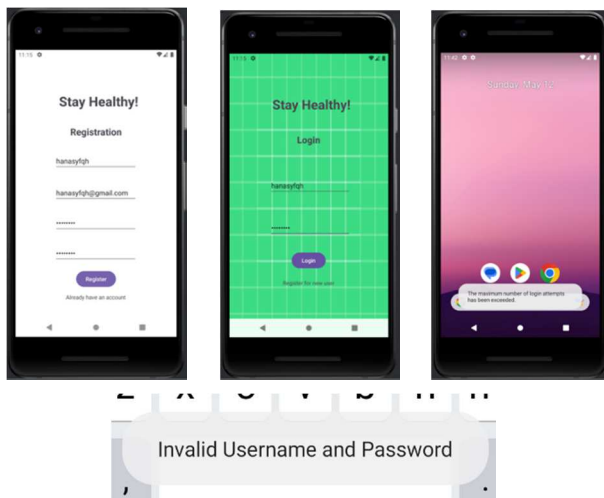
V. IMPLEMENTATION

In this section, we discuss the implementation of the application prototype by using Android Studio as our development environment for designing and developing a secure healthcare application that helps users to utilize healthcare app without worry about the leak of data they store in the app. Android Studio is the official IDE for Android app development, offering a user-friendly interface, code editor, layout editor, build system, emulator, debugging tools, and integration with the Android SDK. It supports multiple programming languages, including Java and Kotlin, and allows developers to design, build, test, and debug Android applications. The layout editor allows developers to visually design the user interface, and the build system uses Gradle for compiling source code and managing dependencies. Android Studio also supports version control systems like Git, enabling collaboration and management of code repositories.

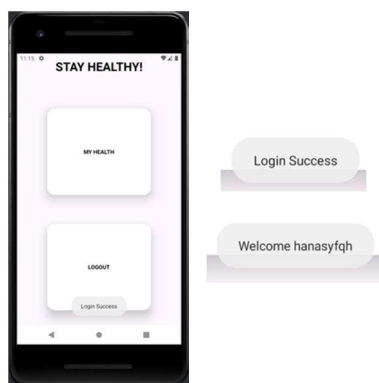
A. Android Application Of Secure Healthcare App

In order to have a deeper comprehension of its operation, let us examine its user interface, as illustrated in Figures 6. Here, we highlight all of its many components

by placing them in a more logical order. The primary features of the apps are the registration page, login page, and health record pages. This Healthcare app name is Stay Healthy!



(a)



(b)

Figure 6: (a) Invalid username or password, (b) Login Success

- **Registration:** Register the information of the user so that later on, user do not need to fill in everytime they want to use the Stay Healthy! app. The information later will be stored in the database.
- **Login:** After user register him/herself in the registration page, they can login anytime they want to use this Stay Healthy! app. We have made a limit attempt to login, which is only 3 attempts to further enhance the security. Lock out user accounts after certain number of incorrect password attempts is a common approach that helps protect against brute force attacks [19]. User can attempt to login only 3 times. After that, the system will exit and display error message saying that user has exceed the maximum number of login attempts. If user enter wrong username or

password less than 3 times, error message will display saying the invalid username and password.

- **Home Page:** As soon as user succesfully logged in to the Stay Healthy!, the home page will be display. It also display short duration messages to the user saying that user has successfully logged in and a short welcoming word to the user. Home next function which are My Health where it will display user health record and Logout section.

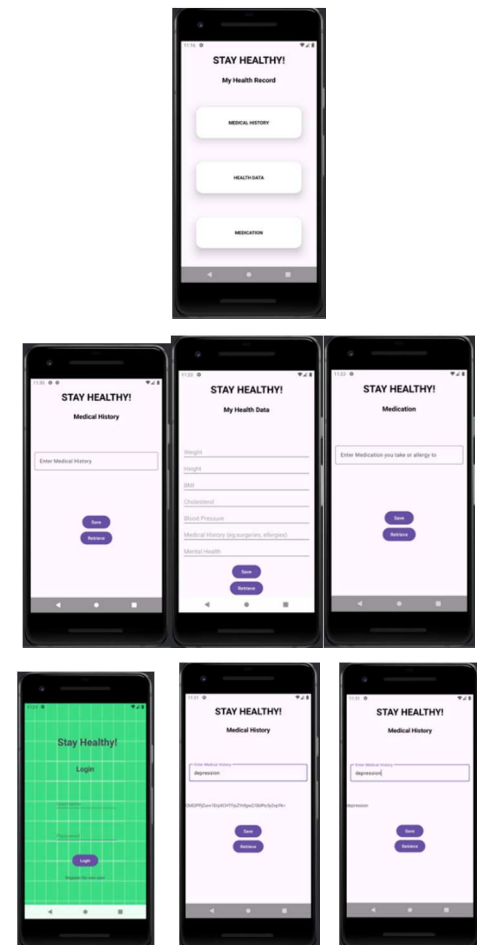


Figure 7: Medical history

- **My Health:** My Health will display another 3 functions which are Medical History Page which stores user's medical history for further reference, Health Data Page which stores user data like weight, height, etc., and Medication Page which stores what medication the user's take and what medication they allergy to. This reference is crucial and sensitive for ensuring their safety life.
- **Logout:** Logout will bring user back to the Login Page. This will ensure security, when the user logout, others cannot enter and cannot access the data.

B. Security in the Healthcare App

This part discusses the detail about the encryption for security purpose.

- Security:** We encrypt data using AES and further encrypy the AES key using ECC. The figure above depicts the security part resides in the Stay Healthy! app. This is the example from Medical History page. When user enter the data and click save button, data will be encrypted. In the real market, the encrypted text will not be displayed to enter safety but in this Stay Healthy! app we want to depict how the text is securely stored. Similarly, the retrieve button also will not be displayed in the real market to ensure safety. But in this paper, we want to depict that we successfully retrieve the data back after we encrypt.

VI. PERFORMANCE ANALYSIS

In this work, the performance of the proposed solution is evaluated by analyzing the power consumption using Activity Monitor and monitoring CPU time. By measuring energy consumption, we gain insight into the efficiency of our application's tasks during encryption, while monitoring CPU time allows us to assess the processing efficiency of this tasks by the CPU. When we layered ECC with AES, we can improve the computational power and CPU time because of the ECC's efficient key management. This can be proven when we real-time monitor energy consumption and CPU time while we run the encryption process in the app. We compare when encrypt data without layering with ECC and with layering ECC to encrypt AES key. The figure 7 and 8 below show the data we collect.

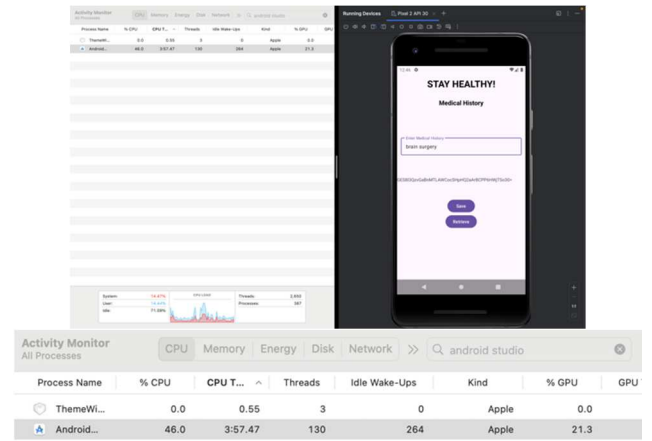
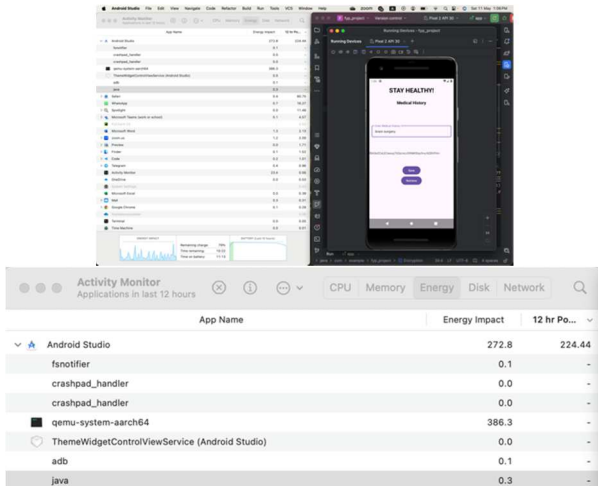


Figure 8: Encryption of Data using AES only without ECC

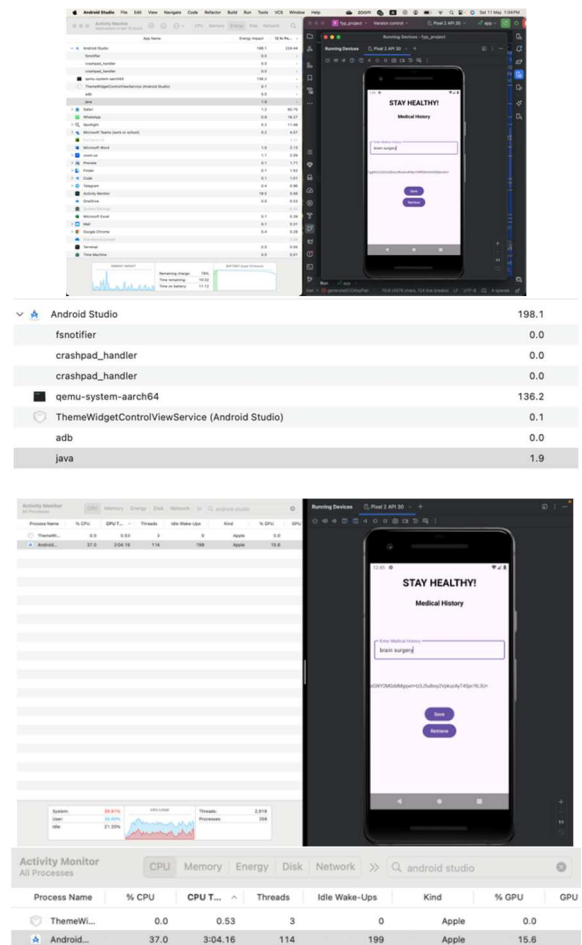


Figure 9: Encryption of Data with AES and Key Encryption with ECC

Hence, it is proven that the performance of the proposed scheme is improved because the energy impact during encryption data using AES only without ECC is 272.8 while encryption of data AES and key encryption with ECC is 198.1. Similar for the CPU time, encryption without ECC is 46.0 while using ECC is 37.0 only.

VII. CONCLUSION

In this paper, we aimed to propose a solution to improve the security of mobile app which is healthcare app as healthcare industry is the most is the biggest and most rapidly expanding sector globally [2]. There are many solutions and methods proposed by other researchers to ensure the security for mobile app. However, there is still issue related to network and data security. Therefore, constant improvement is a must from time to time. We have investigated the current security measure uses to secure mobile apps like cryptography and Intrusion Detection System (IDS). We also identified the limitations and shortcomings of the existing research that open room for much more improvement. Additionally, this paper introduces a method that employs AES for encrypting data and ECC for encrypting keys. This combination offers a secure, established, and computationally efficient algorithm. AES ensures strong encryption for data, while ECC effectively manages key encryption, resulting in a robust and efficient security approach. We managed to study and compare the performance between the proposed algorithm to the previous work algorithm used to secure mobile apps, and we can conclude that all techniques have their own strengths. For future improvement, C++ encryption code optimization and blockchain integration can be implemented to enhance data confidentiality.

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